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The speed of mouse-click as a measure of anxiety during human-computer interaction

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Abstract. The monitoring of the human-computer interaction process is one of the essential aspects in the evaluation and enhancement of both task and affective outcome of human-computer interaction. However, although objective measures exist for task outcome, most affective measures are subjective. This study represented an investigation into the speed of mouse-click as a possible measure in human-computer interaction, and was based principally on the suggestions that a relationship exists between stress and motor activities involved in the operation of the fingers. Two groups of 30 subjects were exposed to different sets of human-computer interaction conditions, and the speed of mouse-click and state anxiety were examined. No correlation was found between the speed of mouse-click and state anxiety. However, a significant difference was found in the speed of mouse-click between the groups and the different human-computer interaction situations. The implication of these findings and the possible advantages of using the computer mouse to collect data relating to the computer user's covert state during human-computer interaction are discussed.

1. Introduction

Although the need for objective means of monitoring the user's affective condition during human-computer interaction should be a central area of research in the field of human-computer interaction, it is in that of Affective Computing that it is presently seen as an important goal. Affective Computing is the area of computing related to the realization of an adaptive computer, and the general driving belief is that a computer system that can recognize the user's emotional state is better placed to provide the assistance likely to lead to better improvement in human-computer interaction (Dryer and Horowitz 1997). Affective Computing also covers the ability of the computer to express some sort of emotion, as well as the related technological principles. However, the focus of the present study was not Affective Computing, but a method that might find a role in the monitoring of emotional state in Affective Computing and human-computer interaction in general.

One way Affective Computing has been described is that it is computing that is associated with emotions (Picard 1997). At the heart of it, therefore, are the monitoring and the recognition of signals related to the user's emotional state. Such signals include facial expression (Ekman 1993), vocal intonation (van Bezooyen 1984), and physiological responses (Picard 1997). It is expected that these signals can be combined to assist in the recognition of emotional states. For example, it has been suggested that physiological measures such as galvanic skin resistance (GSR), heart rate, skin temperature, and general somatic activity are sufficient to determine reliably a person's emotional state (Ark *et al*. 1999).

Presently, a number of different methods exist for monitoring these signals during human-computer interaction. For example, the Emotional Mouse (Ark *et al*. 1999) can measure a number of physiological signals to assess the user's emotional state, and the Sentic Mouse (Kirsch 1997) the finger pressure. However, most of these methods either require the attachment of extra visible devices to the computer user or to the mouse, making the process cumbersome and intrusive, and possibly discouraging to practical application. The present study represented an investigation into a non-invasive method of measuring anxiety during human-computer interaction, based on how fast the computer mouse-button is clicked. It was predicated on the rationale that if the amount of force used to press a button is related to muscle tension, as measured through electromyogram (Suzuki 1995), and muscle tension correlates to psychological stress (Malmo 1975), then it is feasible that how fast the mouse-button is clicked will be influenced by the individual's affective condition.

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A mouse-based method of capturing data relating to the user's affective state during human-computer interaction can offer several advantages from the point of view of practicability. In the main:

- The method requires no attachment to the user or the computer mouse;
- The computer mouse is part of a standard computer setup and therefore data can be collected without the knowledge of the user, thereby possibly minimizing the faking of the data;
- The computer mouse is used to interact with most computer programs and therefore no extra activity is required by the user to provide the data;
- The method can be combined with other methods to further improve emotional state estimate of the computer user.

2. Background

One of the most common ways of measuring anxiety is through subjective measures. While many of these, such as State-Trait Anxiety Inventory (STAI), are well-established and well-proven, subjective measures in general have been reported to be unreliable. Edwards (1957) long since criticised most subjective measures as simply reflecting the extent to which a person describes himself/herself in a socially desirable or undesirable manner, implying that too much is left in the control of the testee, rendering them open to falsification and inaccuracy (Spielberger 1972) Even if their reliability were not debatable, subjective measures can be intrusive, in that the testee is usually required to interrupt the task at hand to attend to them. On the other hand, non-subjective measures of anxiety, such physiological and behavioural measures, have equally been criticized as unreliable, in that they can also be indices for covert activities that are not anxiety-related (Gatchel 1979, Holroyd and Appel 1980). However, given all these, and possible lack of correlation between different measures (Phillips *et al*. 1972), it is still suggested that the more the number of measures used the better (Gatchel 1979, Whittenberger 1979, Nagane 1990).

The need to monitor anxiety during human-computer interaction is supported by many studies which have found that anxiety can affect the performance of computer-related tasks (O'Neil *et al*. 1969, O'Neil and Richardson 1977, Farina *et al*. 1991, Maurer and Simonson 1994, Cooper and Stone 1996). Monitoring might help, for example, in adapting the computer's behaviour accordingly to benefit the human-computer interaction situation. Naturally, there is equally the need for the monitoring to be in a manner that is not invasive, interruptive, or detrimental to the task being performed. Presently, the computer mouse represents one of the best vehicles to accomplish this, particularly because it is one of the devices with which the user mostly interacts with the computer and provides almost continuous data about the interaction.

One definition of anxiety is that it is a component of stress, while stress has been defined as a multidimensional response to demands perceived as excessive by the individual. This multidimensional response can be physiological, behavioural, psychological, or a combination of these. Physiological response relates, for example, to increased heart rate and electromyogram (EMG). The behavioural response is characterised by aggressiveness or passivity, or anger and withdrawal. The psychological response has been described as divisible into two parts: cognitive and emotional. Whereas the cognitive response concerns the evaluation of the situation as threatening, the emotional part is related to states such as anxiety (Vasilaki 1992). The presence of anxiety therefore is a possible indicator of some sort of stress. One commonly recognized possible source of stress – and therefore anxiety – is evaluation, such as is inherent in learning (Hembree 1988, Gonzalez 1995, Spielberger and Vagg 1995).

Several explanations have been suggested to account for why anxiety might affect motor action and why it is a possible index of anxiety. One is physiological in nature and derives from the notion that stress, of which anxiety is a suggested component (Vasilaki 1992), is accompanied by various physiological changes, such as alterations in heart rate (Nagane 1990, Morse 1993), the brain's electrical activity (Hardt and Kamiya 1978, Davidson 1992, Petruzello and Landers 1994, Field *et al*. 1998), and muscle tension (Malmo 1975, Wærsted *et al*. 1991, 1996, Lundberg *et al*. 1994, Larsson *et al*. 1995). These changes consequently would be expected to affect the motor activities involved in the operation of the finger. Thus, for example, increased muscle tension which occurs during increased stress conditions would be expected to affect the motor activity involved in the pressing of the mouse button, resulting in an increase or a decrease in how fast the mouse-button is depressed. After all, pen pressure during writing has been correlated to stress (Keinan and Eilat-Greenberg 1993).

Another possible explanation for the relationship between motor actions and anxiety, which is cognitive in nature, is based on the suggestion that exposure to stress can cause the individual to divide attention between the threatening stimuli and the task being performed (Wine 1971). Thus, in the case of pressing the mouse button, less or more attention might be paid to the task of pressing the mouse button, resulting in the button being pressed faster or more slowly.

In particular, a number of studies that link muscle tension with the pressing of the mouse-button, or the computer keyboard, provide important support for the proposition that stress or anxiety might affect speed of mouse-click during human-computer interaction. Malmo (1975) investigated anxiety and muscle tension by using anxiety-evoking stimuli that ranged from loud noise to thermal pain, and found increased anxiety to correlate to muscle tension. Budzynski and Stoyva (1969) linked anxiety with muscle tension through inducing relaxation in their subjects, and reported a correlation between lower levels of muscle tension and deeper level of relaxation. Jansen and Hoffman (1965) noted that under conditions of stress there were changes in the motor actions of the fingers, such that the size of handwriting and the amount of pencil/pen pressure on paper were affected. Kao *et al.* (1986) correlated handwriting pressure and task complexity, suggesting a link between inability to cope, or difficulty in coping, and the motor actions of the fingers. Inability to cope, or difficulty in coping, has been associated with stress. Other studies (Suzuki 1995, Jensen *et al.* 1998, Karlqvit *et al.* 1998, Birch *et al.* 2000, Johnson *et al.* 2000) suggest a link between muscle tension and the finger pressure used to operate either the computer mouse-button or keyboard.

However, none of the aforementioned studies have directly linked emotional state to the speed of mouse-click during human-computer interaction. The present study set out to investigate the possibility of this relationship. It was not possible to measure the force of mouse-click, due to lack of appropriate apparatus. However, it is feasible that there is a relationship between this description of how the mouse-button is operated and how fast it is clicked. Future studies might investigate this possible relationship. It should also be noted at this point that although muscle tension has been associated with cognitive activities (Wærsted *et al.* 1991, 1996, Lundberg *et al.* 1994, Larsson *et al.* 1995), to simplify the experimental design and minimize the number of possible confounding factors, this relationship was not specifically investigated in the present study. The same rationale applied to the decision not to distinguish between factors such as sexes. Consequently, the main hypothesis investigated in the present study was that there would be a correlation between how fast the mouse-button was clicked and anxiety level.

3. Methodology

3.1. *Subjects*

Two groups of 30 computing undergraduate students participated in the study. Their ages ranged from 18 to 41. They have all been using computers for more than a year and were right-handed. They were recruited through email requesting for volunteers.

3.2. *Apparatus*

3.2.1. *State-trait anxiety inventory (STAI)*: The STAI is an instrument developed by Spielberger *et al.* (1970), and was designed for the purpose of assessing individuals' chronic and state anxiety. The instrument consists of two scales, each comprising 20 items. The A-State scale measures how an individual feels at a particular moment, while the A-Trait how he or she feels generally. Only those items measuring state anxiety were used in the present study. The STAI instrument was used as a measure of anxiety in the study due to its proven reliability and validity. Reliability of the instrument as assessed via a test-retest procedure produce alpha coefficients ranging from 0.73 to 0.86 for college students (Spielberger 1983). Construct validity of the state anxiety scale has been supported by comparison of the mean scores of military recruits tested just prior to beginning a highly stressful training programme, with college and high school students of an approximately similar age who were tested under relatively non-stressful conditions (Spielberger *et al.* 1970).

3.2.2. *Mouse data*: To measure how fast the mouse-button during computer mouse operation, a conventional mouse (Microsoft Corporation) was modified both mechanically and electronically. Due to the way the micro-switch in the mouse works, it was possible to measure how fast the mouse-button travels from its resting (up) position to its depressed (down) position. How fast the button was clicked in this case was the distance between the initial and final points divided by the time taken by the connector to travel between them when the mouse-button is pressed. Due to patent restrictions, details of the modification could not be given in this paper. The data stream from the mouse was digitized and fed directly to a PC through the RS232 connection. The lowest reading possible was zero and the highest 75. Emphasis on units of measure was considered unnecessary at this stage of investigation. It was adequate simply that the measure was consistent all through the study.

3.2.3. *Computer applications*: Two computer-based learning (CBL) applications were used: one for the experimental condition and the other for the control. The application for the experimental condition comprised a tutorial and a test, and the subject matter was a brief 'History of Human-Computer Interaction'. There

were 14 slides of learning materials in all, and they were programmed such that once a slide was passed, review was not possible. The test comprised 14 multiple-choice questions, each consisting of four options A, B, C, and D. The test condition was considered capable of evoking increased stress, as measured by anxiety level (Spielberger 1966, O'Neil *et al*. 1977, Cooper and Stone 1996). The application therefore provided two conditions (tutorial and test) capable of evoking differing levels of anxiety. The application recorded how fast the mouse-button was clicked each time the mouse was clicked. The mouse was only clicked to advance from one slide to the next both during the tutorial and the test.

The control application comprised two tutorials, but without a test. Each tutorial consisted of 14 slides of learning materials on 'History of Human-Computer Interaction'. As with the experimental application, the mouse was only clicked to advance from one slide to the next.

4. Procedure

The participants were allocated randomly to either the experimental or control group before their arrival. On arrival, each participant was seated in front of the computer in a quiet room, the general nature of the experiment was explained, and consent taken. The monitor was place halfway along the length of the worktable and a distance of 0.5 meter from the front edge. The height from the worktable surface to the mid-point of the screen was 0.3 meter. To reduce confounding factors that might be due to posture, the chair was adjusted so that the arm was supported against the worktable. The mouse was fixed and its centre placed at a point 30 cm from the front edge of the worktable and 40 cm from the middle of length of the worktable, and to the right. This mouse position represented one found to minimize muscle load and one that is generally perceived to be the most comfortable by computer mouse operators (Karlqvist *et al*. 1998).

For the experimental group, each participant was informed that the session comprised first the tutorial and then the test. During the tutorial condition, the participant went through 14 slides of material. To advance from one slide to another, the participant clicked the mouse, at which point the mouse-click speed (MCS) was recorded. Overall, 14 clicks were taken, one click for advancing each slide. Immediately after the tutorial, participants were asked to complete the STAI state anxiety measure based on how they felt during the tutorial. After this, the participant immediately went through the test that also comprised 14 slides. Each slide contained a question and four answers. The participant was required to enter his/her choice of answer (i.e., A, B, C, or D), using the computer keyboard, and then clicking the mouse to advance to the next slide. Again, 14 clicks were recorded. After the test, another STAI state anxiety measure was administered based on how the participant felt during the test session. The time taken to complete each of the sessions was also recorded.

For the control group, each participant was informed that the session comprised simply two tutorials and no test. A similar procedure to that of the experimental condition was then followed to collect the mouse-click data. After the first tutorial, the STAI state anxiety measure was completed, then the second tutorial, and then another STAI. The time taken to complete each session was also recorded.

5. Results

Given the small sample sizes, and the fact that nothing was known about mouse-click speed (MCS) in the population, non-parametric tests were used to analyse the data collected. To investigate if there was a relationship between how fast the mouse-button was clicked (MCS) and the STAI state anxiety score, Spearman's correlation analysis was computed for the data collected from the experimental and control groups. For the experimental group, no significant correlation was found either under the tutorial condition ($rho = -0.128$, $df = 28$, $p < 0.446$) or the test condition ($rho = 0.105$, $df = 28$, $p < 0.579$). Equally, for the control group, no significance was found under the first tutorial condition ($rho = 0.206$, $df = 28$, $p < 0.275$) or the second tutorial condition ($rho = 0.235$, $df = 28$, $p < 0.212$). The hypothesis that how fast the mouse-button was clicked would be related to anxiety was therefore not supported.

To investigate if there were significant differences in MCS between the conditions within each group, Wilcoxon Matched Pairs tests were conducted. Significant difference was found in the experimental group between the tutorial and test sessions ($z = -4.784$, two-tailed, $p < 0.001$), whereas none was found in the control group between the first and the second tutorial sessions ($z = -0.710$, two-tailed, $p < 0.478$). See table 1.

To investigate if there were significant differences in MCS between the two groups, Mann–Whitney *U*-tests were conducted. A significant difference in MCS was found between the tutorial in the experimental group and the first tutorial in the control group ($U = 292.00$, two-tailed, $p < 0.019$). Significant difference was also found in MCS between the test in the experimental

group and the second tutorial in the control group ($U = 4.00$, two-tailed, $p < 0.001$). See table 2.

To investigate the possibility of a relationship between the pace of task completion and the speed of mouse click, Spearman's correlation analysis was computed on the time of completion and the speed of mouse-click from both the experimental and control groups. No correlation was found. However, Wilcoxon Matched Pairs analysis performed on the speed of mouse-click of the participants in the experimental group who spent similar times on both the tutorial and the test showed a significance increase ($z = -3.180$, two-tailed, $p < 0.001$).

No significant differences were found in STAI data between the two groups and between the tutorial and test conditions in the experimental group. Mann–Whitney tests showed no significant difference in STAI scores between the experiment and control groups during the tutorial and the first tutorial ($U = 443.00$, two-tailed, $p < 0.917$), nor during the test and second tutorial ($U = 410.00$, two-tailed, $p < 0.552$). A Wilcoxon test showed no significance difference in STAI scores between either the tutorial and test sessions in the experimental group ($z = -1.660$, two-tailed, $p < 0.097$), or the first and second tutorials in the control group ($z = -0.957$, two-tailed, $p < 0.338$).

6. Discussion

This study indicated no obvious correlation between the speed of mouse-click and state anxiety. This was an unexpected finding, as clicking the mouse-button is a form of motor activity and behaviour expression, and both of these have been associated with anxiety or stress. However, the fact that no correlation was found in this study may not necessarily mean that the pressure used to click the mouse button is not related to anxiety. It is feasible, given the rationale already stated above, that other techniques of measuring finger pressure would reveal different. The present study found only that there was no indication of a relationship between the speed of mouse-click and subjective state anxiety during a learning-based human-computer interaction.

The absence of an evidence of a relationship between anxiety level and the speed of mouse-click may suggest that the significant increase in the speed of mouse-click during the test period in the experimental group was due to other factors. However, in this instance, one of these factors could not be assumed to be time or pace of task completion, since a significant increase was also found in the speed of mouse-click of those who spent similar amounts of time on both the tutorial and the test. In other words, the time interval between each mouse-click did not seem to have had any significant effects on the speed of the mouse-clicks in this study. This may suggest, therefore, that the increase in the speed of mouse-click was due to factors such as cognitive activities relating to the recalling of the required information during the test. That is, the evaluative situation possibly necessitated certain cognitive activities which in turn influenced the speed of mouse-click. Although the present study did not investigate this aspect, future studies may do this and possibly help to more specifically identify the nature of these cognitive activities, and the possible application of their monitoring in effective human-computer interaction, particularly in computer-based learning.

One generalization of the above finding is via deduction that an increase in the intensity of cognitive activity seems to be related to increase in the speed of mouse-click during human-computer interaction. The other finding that could be generalized was the one that the speed of mouse-click in the present study varied significantly with changes in the human-computer interaction situations in the experimental group. This was in sharp contrast to the control group where the human-computer interaction situation was fairly constant and no significant changes were found in the speed of mouse-click. From this observation it may be suggested that how fast the mouse-button is clicked is a behaviour that varies according to the intensity cognitive activities during human-computer interaction, and could have application in the monitoring of the computer user during human-computer interaction.

From a theoretical point of view, the absence of a significant increase in state anxiety during the test session in the experimental group, or a significant difference in anxiety level between the two groups during the test in the experimental group and the second

Table 1. Means and standards deviations of MCS.

	Session 1	Session 2
Control	21.06	21.42
	5.05	**4.69**
Experimental	24.44	36.04
	3.60	**3.46**

Means (Roman Type) and standard deviation (**Boldface Type**)

Table 2. Mean ranks of MCS.

	Session 1	Session 2
Control	25.23	15.63
Experimental	35.77	45.37

tutorial in the control group, contrasted with previous studies (Spielberger 1966, O'Neil *et al*. 1977, Cooper and Stone 1996) that suggested that evaluative situations tend to produce higher state anxiety. One explanation for this contradiction could be that the participants in the present study did not perceive the evaluative situation to be real enough, and therefore did not behave like they would normally do in real and future-determining evaluative situations.

In conclusion, although no correlation was found between the speed of mouse-click and anxiety, the rest of the findings in the study suggested a link between the speed of mouse-click and the nature of human-computer interaction. In particular, the findings provided inducement to further look in the direction of the mouse, and perhaps other interaction devices, for means of monitoring data about the computer user which can be used towards effective human-computer interaction. Furthermore, as implied earlier, although no correlation was found between the speed of mouse-click and anxiety in the present study, this does not necessarily mean that there is no relationship between the computer user's finger operation and anxiety. There is still, through logical deduction, a strong argument for this. Therefore, future studies should investigate this relationship using other methods of finger pressure measurement.

References

Ark, W., Dryer, D. C. and Lu, D. J. 1999, The emotional mouse. http://www.almaden.ibm.com/cs/blueeyes/mouse2.html

Birch, L., Luul-Kristensen, B., Jensen, C., Finsen, L. and Christensen, H. 2000, Acute response to precision, time pressure and mental demand during simulated computer work. *Scandinavian Journal of Environmental Health*, **26**(4), 299 – 305.

Budzynski, T. H. and Stoyva, T. M. 1969, An instrument for producing deep muscle relaxation by means of analogue information feedback. *Journal of Applied Behavior Analysis*, **2,** 231 – 237.

Cooper, J. and Stone, J. 1996, Gender, computer-assisted learning, and anxiety: With a little help from a friend. *Journal of Educational Computing Research*, **15**(1), 67 – 91.

Davidson, R. J. 1992, Anterior cerebral asymmetry and the nature of emotion. *Brain and Cognition*, **20,** 125 – 151.

Dryer, D. C. and Horowitz, L. M. 1997, When do opposites attract? Interpersonal complementarity versus similarity. *Journal of Personality and Social Psychology*, **72,** 592 – 603.

Edwards, A. L. 1957, The Social Desirability Variable in Personality Assessment and Research (New York: Dryden Press).

Ekman, P. 1993, Facial expression and emotion. *American Psychologist*, **48**(4), 384 – 392.

Farina, F., Arce, R., Sobral, J. and Carames, R. 1991, Predictors of anxiety towards computers. *Computers in Human Behavior*, **7,** 263 – 267.

Field, T., Martinez, A., Nawrocki, T., Pickens, J., Fox, N. A. and Schanberg, S. 1998, Music shifts frontal EEG in depressed adolescents. *Adolescence*, **33**(129), 109 – 116.

Fox, N. A. and Davidson, R. J. 1987, Electroencephalogram asymmetry in response to the approach of a stranger and material separation in 10-month-old infants. *Developmental Psychology*, **23,** 233 – 240.

Gatchel, R. J. 1979, Biofeedback and the treatment of fear and anxiety. In R. J. Gatchel and K. P. Price (eds) *Clinical Applications of Biofeedback: Appraisal and Status* (New York: Pergamon Press).

Gonzalez, H. P. 1995, Systematic desensitization, study skills counselling, and anxiety-coping training in the treatment of test anxiety. In C. D. Spielberger and P. R. Vagg (eds) *Test Anxiety: Theory, Assessment, and Treatment* (Washington, DC: Taylor & Francis).

Hardt, J. V. and Kamiya, J. 1978, Anxiety change through electroencephalographic alpha feedback seen only in high anxiety subjects. *Science*, **210,** 79 – 81.

Hembree, R. 1988, Correlates, causes, effects, and treatment of test anxiety. *Review of Educational Research*, **58**(1), 47 – 77.

Holroyd, A. H. and Appel, M. A. 1980, Test anxiety and physiological responding. In I. G. Sarason (ed) *Test Anxiety: Theory, Research and Applications* (Hillsdale, New Jersey: Lawrence Erlbaum).

Jansen, G. and Hoffman, H. 1965, Changes in fine motor movements and annoyance levels due to noise and their relation to certain dimensions of personality. *Zeitschrift fuer Experimentelle und Angewndte Psychologie*, **12,** 594 – 613.

Jensen, C., Borg, V., Finsen, L., Hansen, K., Juul-Kristensen, B. and Christensen, H. 1998, Job demands, muscle activity and musculoskeletal symptoms in relation to work with the computer mouse. *Scandinavian Journal of Environmental Health*, **24**(5), 418 – 424.

Johnson, P. W., Hagberg, M., Hjelm, E. W. and Rempel, D. 2000, Measuring and characterizing force exposures during computer mouse use. *Scandinavian Journal of Environmental Health*, **26**(5), 398 – 405.

Karlqvist, L. K., Bernmark, E., Ekenvall, L., Hagberg, M., Isaksson, A. and Rosto, T. 1998, Computer mouse position as a determinant of posture, muscular load and perceived exertion. *Scandinavian Journal of Environmental Health*, **24**(1), 62 – 73.

Kao, H. S. R., Mak, P. H. and Lam, P. W. 1986, Handwriting pressure: Effects of task complexity, control mode and orthographic difference. In H. S. R. Kao, G. P. van Galen and R. Hoosain (eds), *Graphonomics: Contemporary research in handwriting* (North-Holland, Amsterdam).

Keinan, G. and Eilat-Greenberg, S. 1993, Can stress be measured by handwriting analysis? The effectiveness of the analytic method. *Applied Psychology: An International Review*, **42**(2), 153 – 170.

Kirsch, D. 1997, The sentic mouse: developing a tool for measuring emotional valence. http://www.media.mit.edu/affect/AC_research/projects/sentic_mouse.html

Larsson, S. E., Larsson, R., Zhang, Q., Cai, H. and Öberg, P. Å. 1995, Effects of psychophysiological stress on trapezius muscles blood flow and electromyography during static load. *European Journal of Applied Physiology*, **71,** 493 – 498.

Lundberg, U., Kadefors, R., Melin, B., Palmerud, G., Hassmén, P., Engström, M. and Elfsberg Dohns, I. 1994, Psychophysiological stress and EMG activity of the trapezius muscle. *International Journal of Behavioral Medicine*, **1,** 354 – 370.

Malmo, R. B. 1975, *On Emotions, Needs, and Our Archaic Brain* (New York: Holt, Rinehart and Winston, Inc.).

Maurer, M. M. and Simonson, M. R. 1994, The reduction of computer anxiety: Its relation to relaxation training, previous computer coursework, achievement, and need for cognition. *Journal of Research in Computing Education*, **26**(2), 205–219.

Morse, D. R. 1993, Brain wave synchronizers: A review of their stress reduction effects and clinical studies assessed by questionnaire, galvanic skin resistance, pulse rate, saliva and electroencephalograph. *Stress Medicine*, **9,** 111–126.

Nagane, M. 1990, Development of psychological and physiological sensitivity indices to stress based on state anxiety and heart rate. *Perceptual and Motor Skills*, **70,** 611–614.

O'Neil, H. F. Jr., Spielberger, C. D. and Hansen, D. N. 1969, Effects of state anxiety and task difficulty on computer-assisted learning. *Journal of Educational Psychology*, **60,** 343–350.

O'Neil, H. F., Jr. and Richardson, F. C. 1977, Anxiety and Learning in Computer-Based Learning Environments: An Overview. In J. E. Sieber, H. F. O'Niel, Jr. and S. Tobias (eds), *Anxiety, Learning, and Instruction* (Hillsdale, New Jersey: Lawrence Erlbaum).

Petruzzello, S. J. and Landers, D. M. 1994, State anxiety reduction and exercise. Does hemispheric activation reflect such changes? *Medical Science Sports Exercise*, **26**(8), 1028–1035.

Phillips, B. N., Martin, R. P. and Meyers, J. 1972, Interventions in relation to anxiety in school. In C. D. Spielberger (ed), *Anxiety: Current Trends in Theory and Research*. Vol. 2 (New York: Academic Press).

Picard, R. 1997, Affective Computing (Cambridge: The MIT Press).

Spielberger, C. D. 1966, The effects of anxiety on complex learning and academic achievement. In C. D. Spielberger (ed), *Anxiety and Behavior* (New York: Academic Press).

Spielberger, C. D., Gorsuch, R. L. and Lushene, R. E. 1970, *The State-Trait Anxiety Inventory (STAI), Test Manual for Form X* (Palo Alto, California: Consulting Psychologists Press).

Spielberger, C. D. 1972, Anxiety as an Emotional State. In C. D. Spielberger, (ed) *Anxiety: Current Trends in Theory and Research*, Vol. 1 (New York: Academic Press).

Spielberger, C. D., Gorsuch, R. L., Lushene, R., Vagg, P. R., and Jacobs, G. A. 1983, State-Trait Inventory (Form Y) (CA: Consulting Psychologists Press).

Spielberger, C. D. and Vagg, P. R. 1995, Test Anxiety: Theory, Assessment, and Treatment (Washington, DC: Taylor and Francis).

Suzuki, K. 1995, On a simple method to measure the intensity of keystrokes. In Y. Anzai, K. Ogawa and H. Mori (eds) *Symbiosis of Human and Artifact: Human and Social Aspects of Human-Computer Interaction*, *Advances in Human Factors/Ergonomics*, *Vol. 2* (London: Elsevier).

van Bezooyen, R. 1984, Characteristics and recognizability of vocal expressions of emotions (New Jersey: Foris Publications).

Vasilaki, E. 1992, *Stress and anxiety in Scottish and Greek high school pupils* (PhD Thesis: Dundee University).

Wærsted, M., Bjorklund, R. and Westgaard, R. 1991, Shoulder muscle tension induced by two VDU-based tasks of different complexity. *Ergonomics*, **34,** 137–150.

Wærsted, M., Eken, T. and Westgaard, R. H. 1996, Activity of single motor units in attention-demanding tasks: firing pattern in the human trapezius muscle. *European Journal of Applied Physiology*, **72,** 323–329.

Wine, J. 1971, Test anxiety and direction of attention. *Psychological Bulletin*, **76,** 92–104.

Whittenberger, G. 1979, *Correlation of magnitude estimates of state anxiety with heart rate, finger pulse volume, skin conductance, and electromyogram responses for subjects under threat of shock* (PhD Thesis: Florida State University).

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